

影像處理 霍夫直線

教師:許閔傑、蕭兆翔

助教:莊媿涵

影像重建大綱

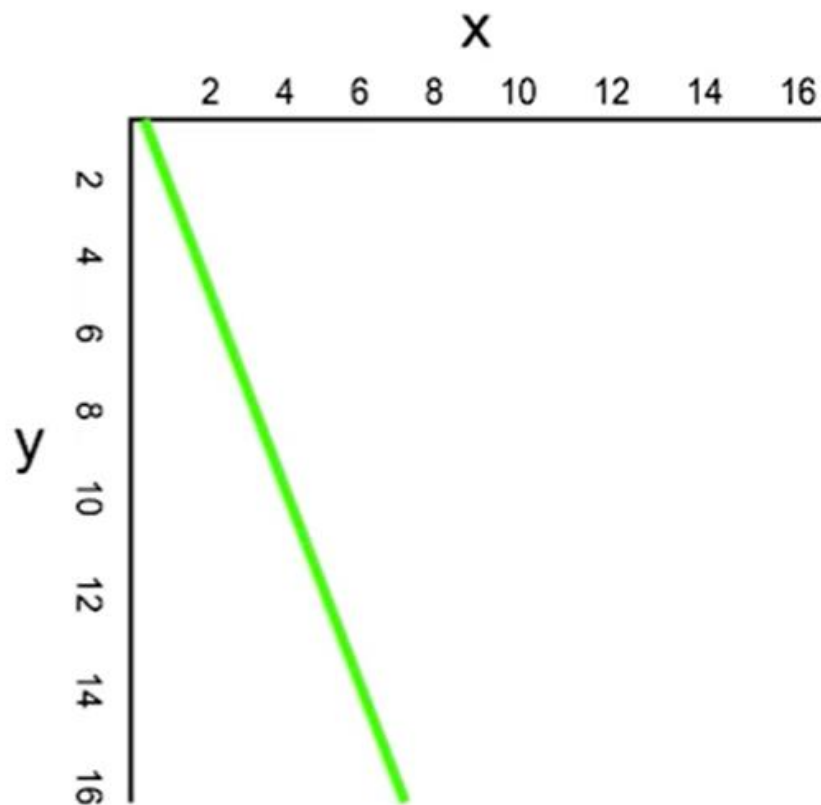
- 霍夫空間
- 霍夫直線轉換
- HOG

霍夫空間

$$y = mx + b$$

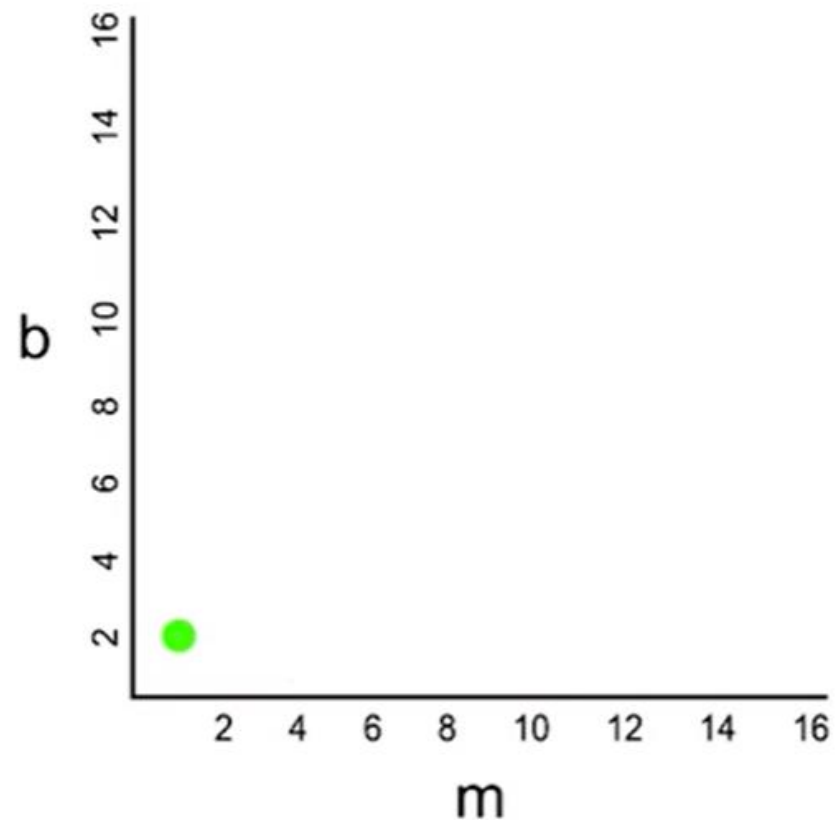
$$y = 2x + 1$$

Cartesian



(2,1)

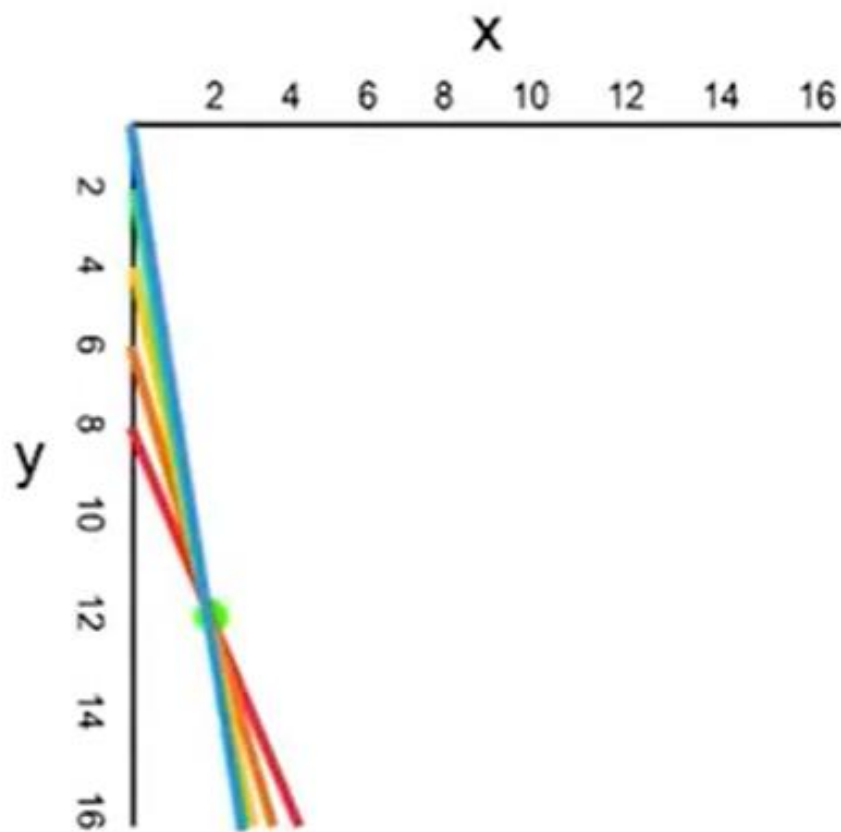
Hough space



霍夫空間

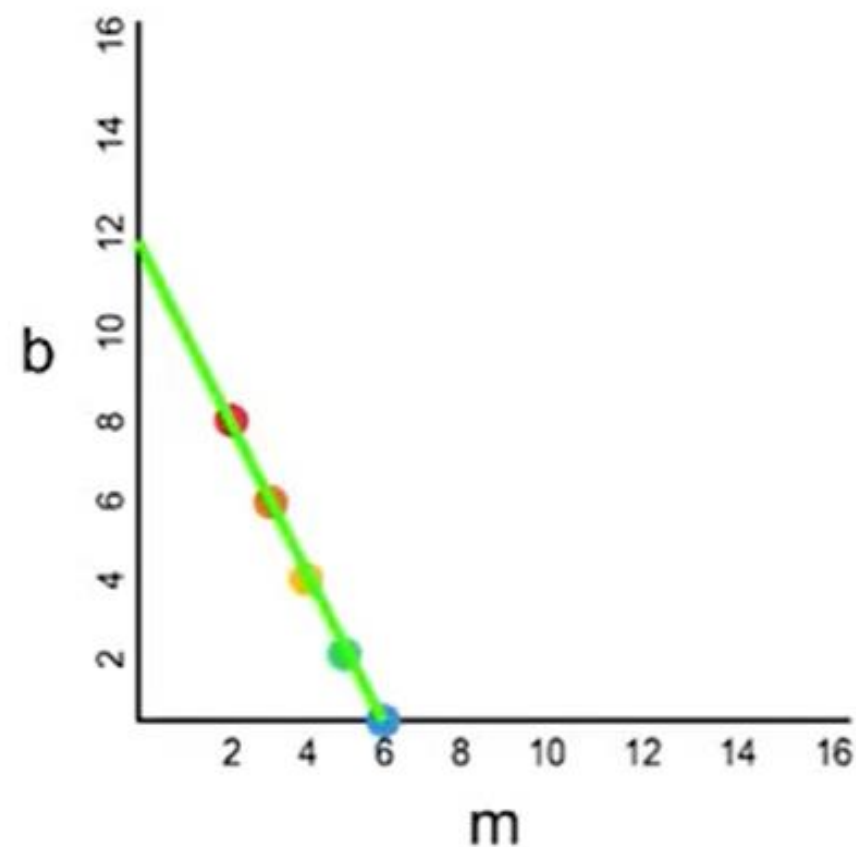
許多線通過一點

Cartesian



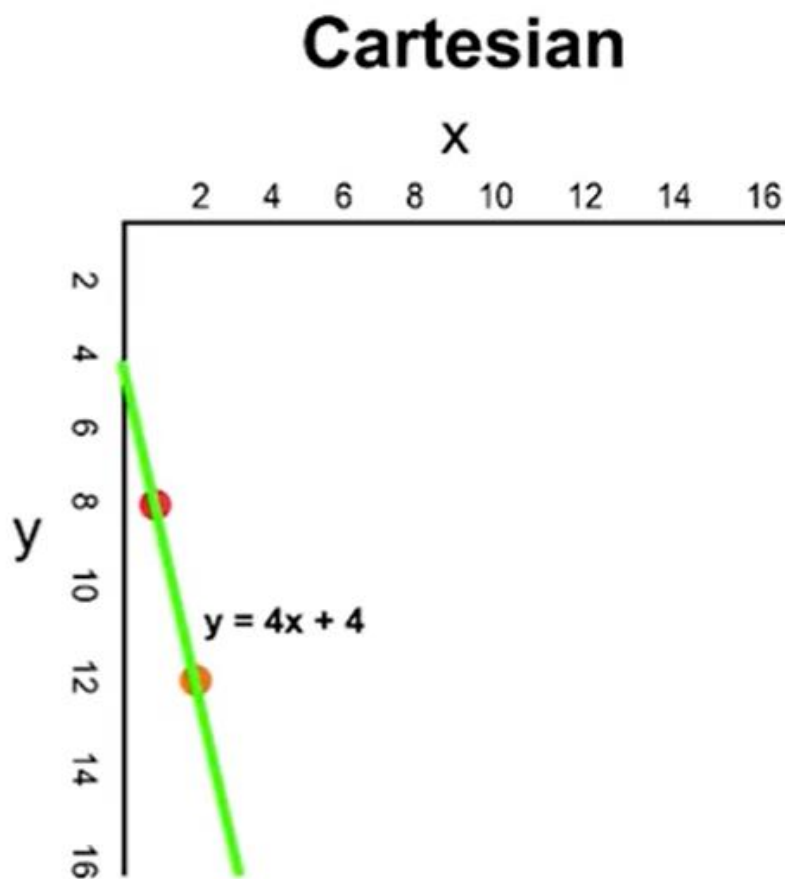
許多點連成一線

Hough space

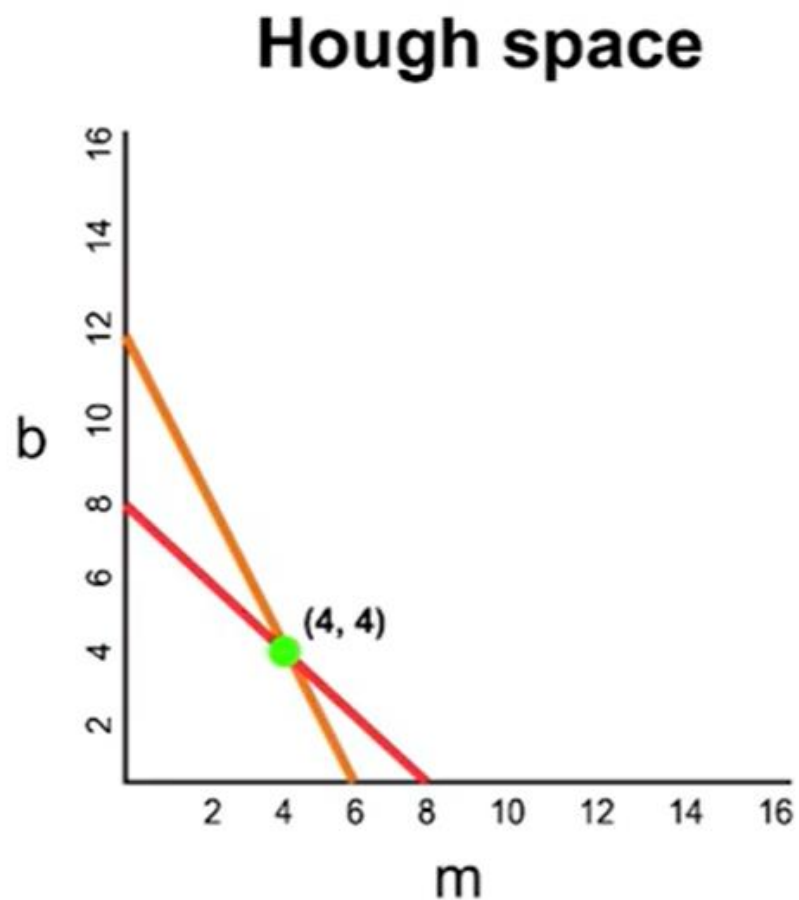


霍夫空間

一線通過許多點

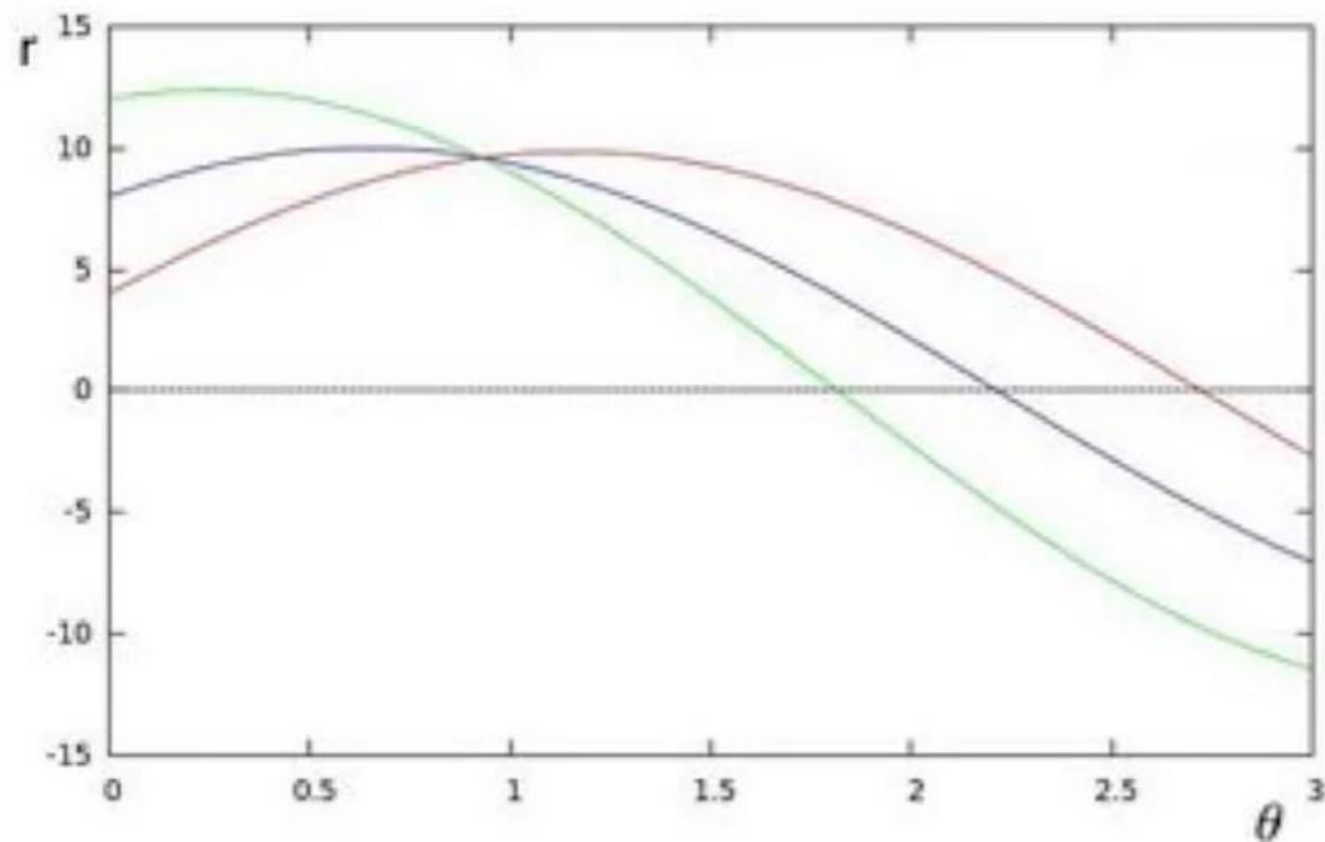
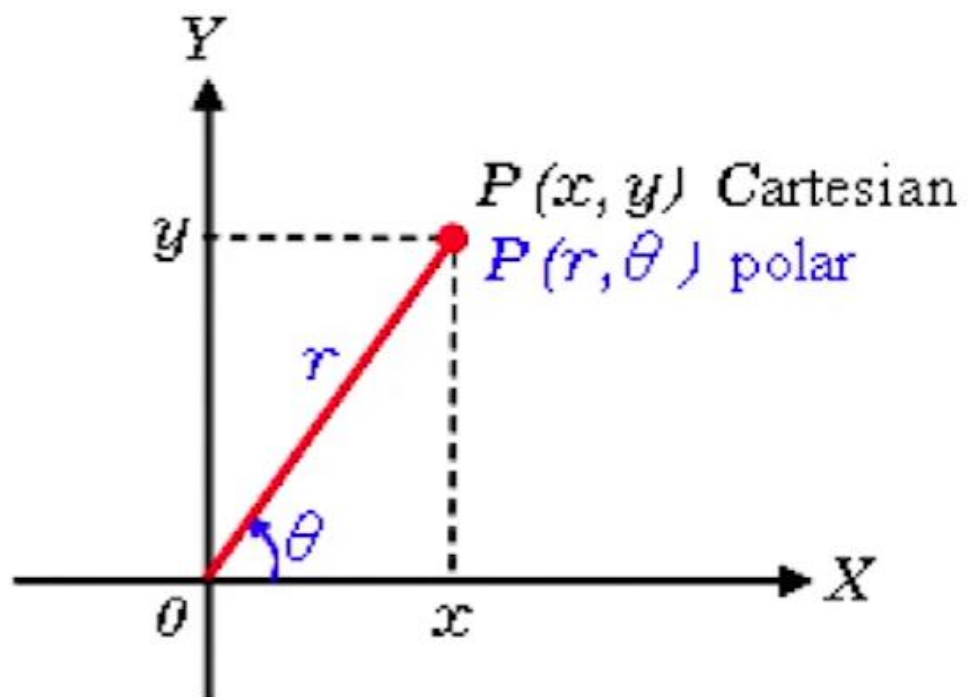


多線通過一點



極座標表示

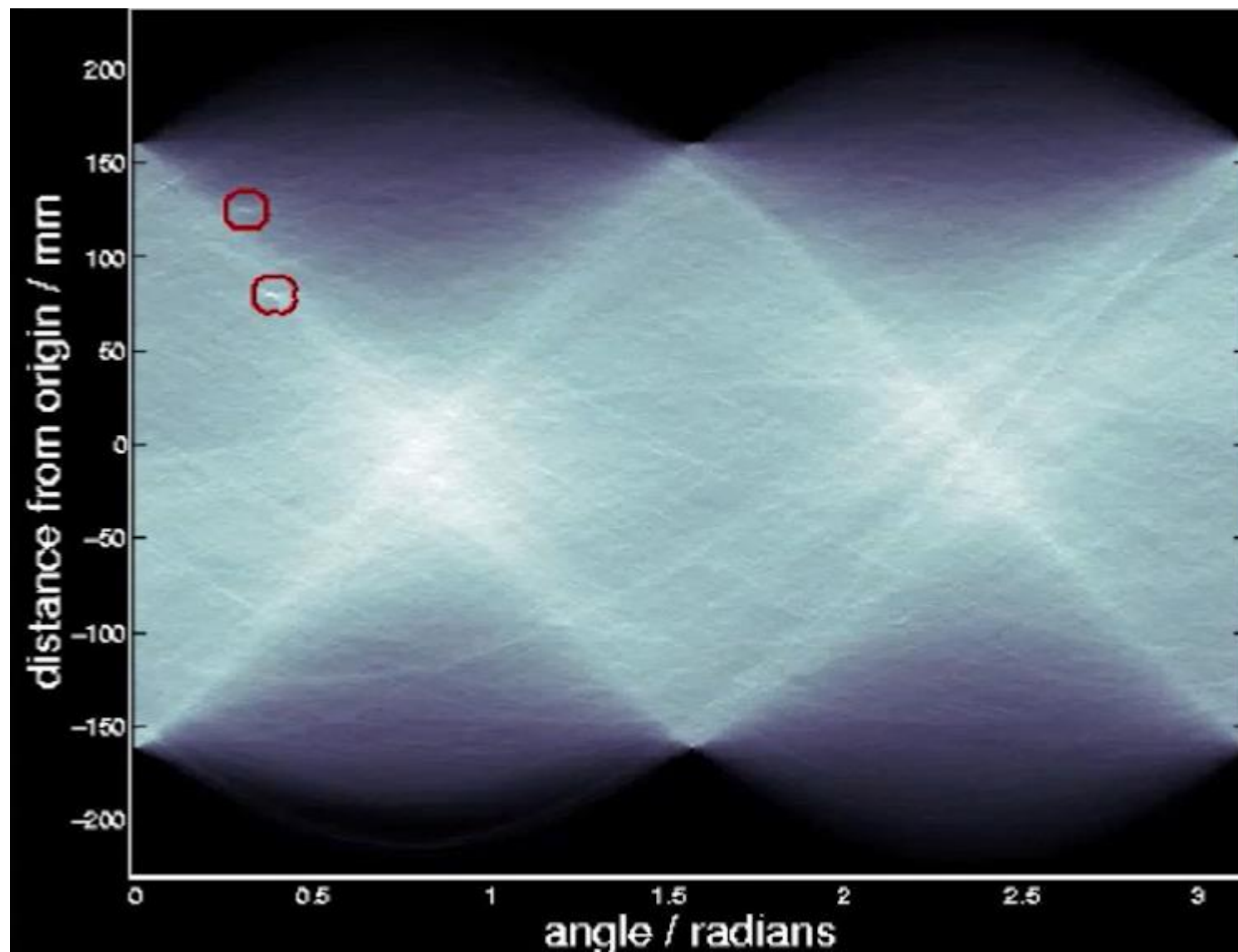
$$r = x \cos \theta + y \sin \theta$$



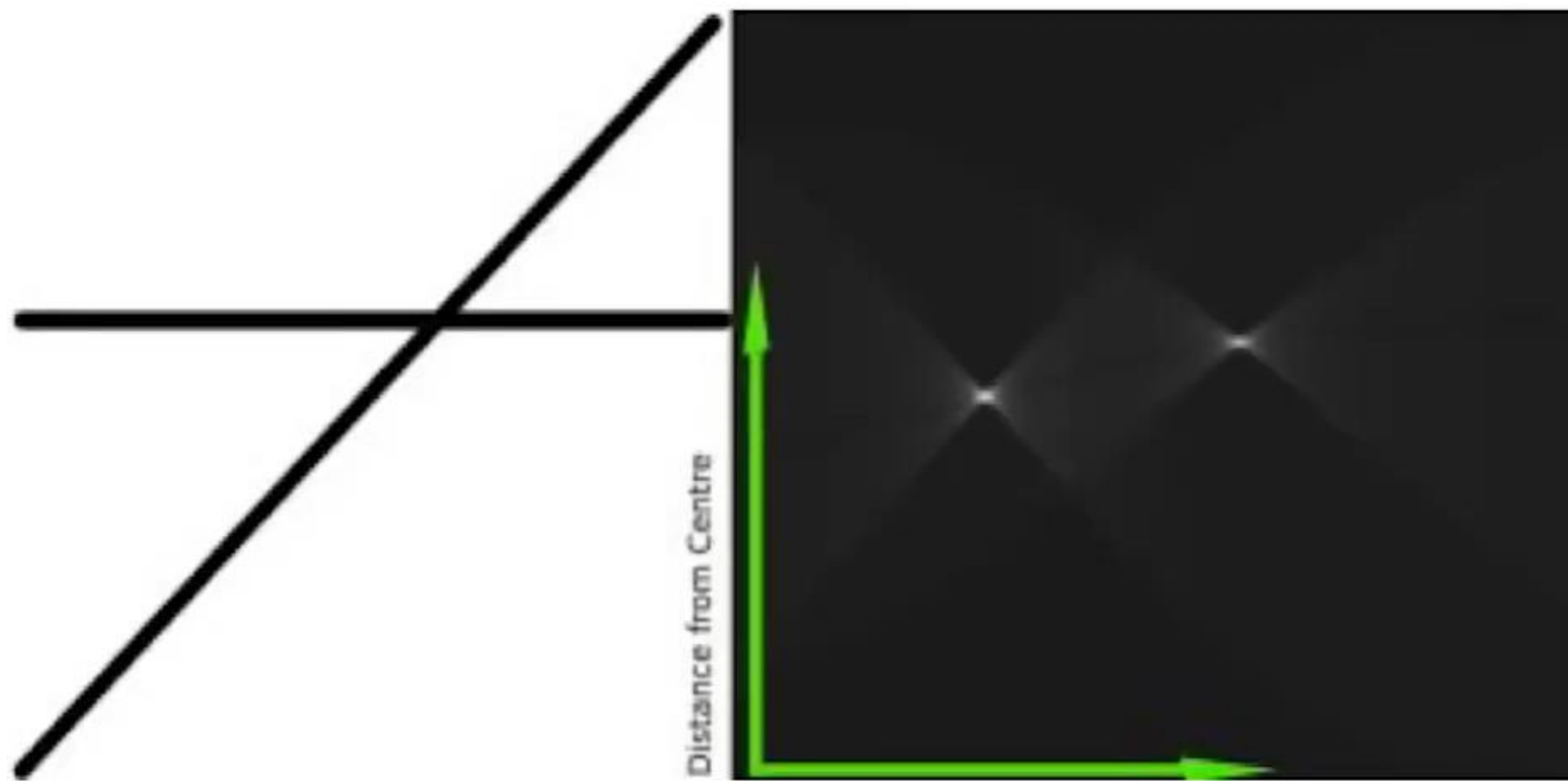
Canny邊緣檢測



真實情況



結果



HOG

<https://www.youtube.com/watch?v=5nZGnYPyKLU&t=16s>

